36.

```c
#include<stdio.h>
int main()
{
    int a, i, j;
    printf("Enter the number of rows:");
        scanf("%d", &a);
    for(i=65;i<65+a;i++)
        {
            for(j=65;j<=i;j++)
                {
                    printf("%c", j);
                }
            printf("\n");
        }
    return 0;
}
```

```
Enter the number of rows:5
A
AB
ABC
ABCD
ABCDE
```

37.

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= i; j++)
        {
            printf("%d", j);
        }
        printf("\n");
    }

    return 0;
}
```

```
    1
   12
  123
 1234
12345
```

38.

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= i; j++)
        {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

```
    *
   **
  ***
 ****
*****
```

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= i; j++)
        {
            printf("%c", 'A' + i - 1);
        }
        printf("\n");
    }

    return 0;
}
```

```
Enter number of rows: 5
    A
   BB
  CCC
 DDDD
EEEEE
```

40.

```c
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
        {
            if (i == j)
                printf("1");
            else
                printf("0");
        }
        printf("\n");
    }

    return 0;
}
```

```
Enter number of rows: 6
100000
010000
001000
000100
000010
000001
```

41.

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= i; j++)
        {
            printf("%d", j);
        }
        for (j = i - 1; j >= 1; j--)
        {
            printf("%d", j);
        }
        printf("\n");
    }

    return 0;
}
```

```
Enter number of rows: 6
     1
    121
   12321
  1234321
 123454321
12345654321
```

```c
int main()
{
    int n, i, j, k;

    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++)
        {
            printf("*");
        }
        printf("\n");
    }

    for (i = n - 1; i >= 1; i--)
    {
        for (k = 1; k <= n - i; k++)
        {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++)
        {
            printf("*");5
        }
        printf("\n");
    }

    return 0;
}
```

```
Enter number of rows for upper half: 5
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

43) PROGRAM TO RED AND ARRAY OF 10 INTEGER AND PRINT SUM OF NUMBERS.

```c
//sum of 10 integers in an array
#include<stdio.h>
int main()
{
    int a[10],i,j,sum=0;
    printf("enter the numbers:");
    for(i=0;i<10;i++)
        {
            scanf("%d", &a[i]);
        }
    for(i=0;i<10;i++)
        {
            sum=sum+a[i];
        }
    printf("\nsum=%d", sum);
return 0;

}
```

```
enter the numbers:25
35
45
55
65
75
85
95
15
5

sum=500
```

44) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO OF ODD AND TOTAL NO. OF EVERY ELEMENT.

```c
//total odd and even numbers
#include<stdio.h>
int main()
{
    int n, i, j, countOdd=0, countEven=0;
    printf("enter the size of array ");
    scanf("%d", &n);
    int a[n];
    printf("Enter the numbers:");
        for(i=0;i<n;i++)
        {
          scanf("%d", &a[i]);
        }
    for(i=0;i<n;i++)
        {
            if(a[i]%2==0)
            {
                countEven+=1;
            }
            else
            {
                countOdd+=1;
            }
        }
    printf("total no: of even and odd = %d ,%d ", countEven , countOdd);
return 0;

}
```

```
enter the size of array 10
Enter the numbers:1
2
3
4
5
6
7
8
9
10
total no: of even and odd = 5 ,5
```

45) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO. OF POSITIVE, NEGATIVE, AND ZERO ELEMENTS.

```c
#include<stdio.h>
int main()
{
    int a[10], i, countPositive=0, countNegative=0, countZero=0;
    printf("Enter the numbers:");
    for(i=0;i<10;i++)
        {
            scanf("%d", &a[i]);
        }
    for(i=0;i<10;i++)
        {
            if(a[i]>0)
            {
                countPositive+=1;
            }
            if(a[i]<0)
            {
                countNegative+=1;
            }
            if(a[i]==0)
            {
                countZero+=1;
            }

        }
    printf("Total positive integers is %d , negative integer is %d, and zero is %d",
        countPositive, countNegative, countZero);
    return 0;
}
```

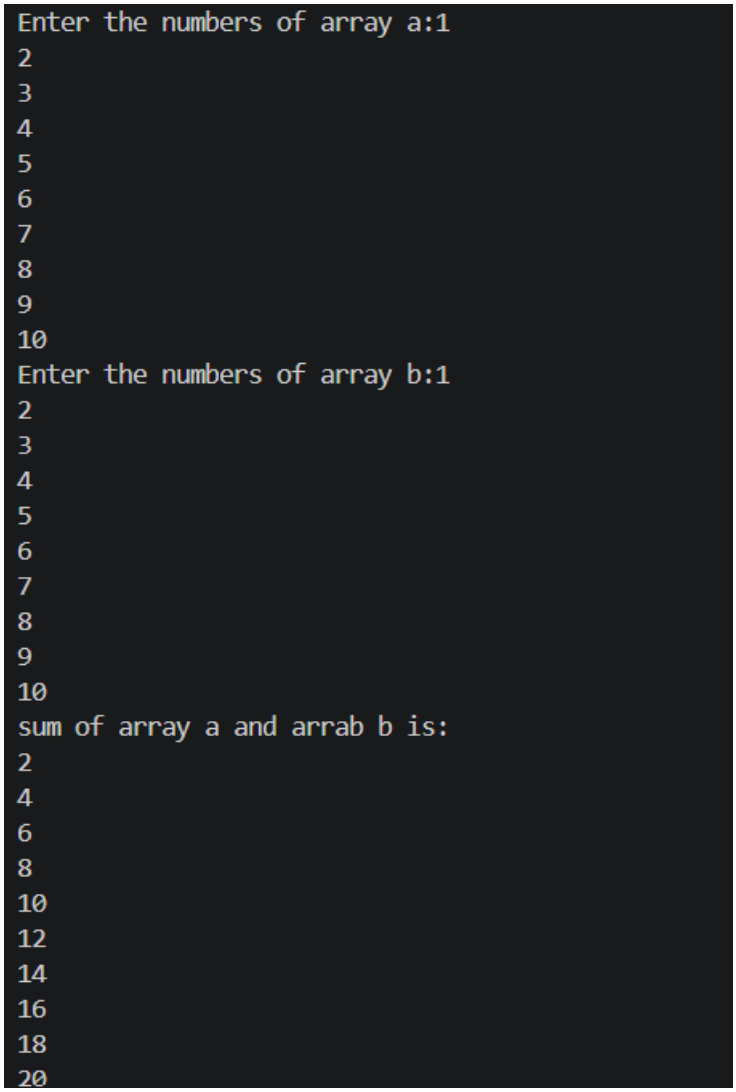
```
Enter the numbers:12
65
-45
-3
0
56
-9
24
0
52
Total positive integers is 5 , negative integer is 3, and zero is 2
```

47) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND STORE ADDATION OF THOSE ARRAYS INTO THIRD.

```c
#include<stdio.h>
int main()
{
    int i, max;
    int a[10];
    printf("Enter the numbers of array a:");
        for(i=0;i<10;i++)
        {
          scanf("%d", &a[i]);
        }
    int b[10];
    printf("Enter the numbers of array b:");
        for(i=0;i<10;i++)
        {
          scanf("\n%d", &b[i]);
        }
    int c[10];

            printf("sum of array a and arrab b is:");
    for(i=0;i<10;i++)
        {
            c[i]=a[i]+b[i];
            printf("\n%d ", c[i]);
        }
return 0;

}
```

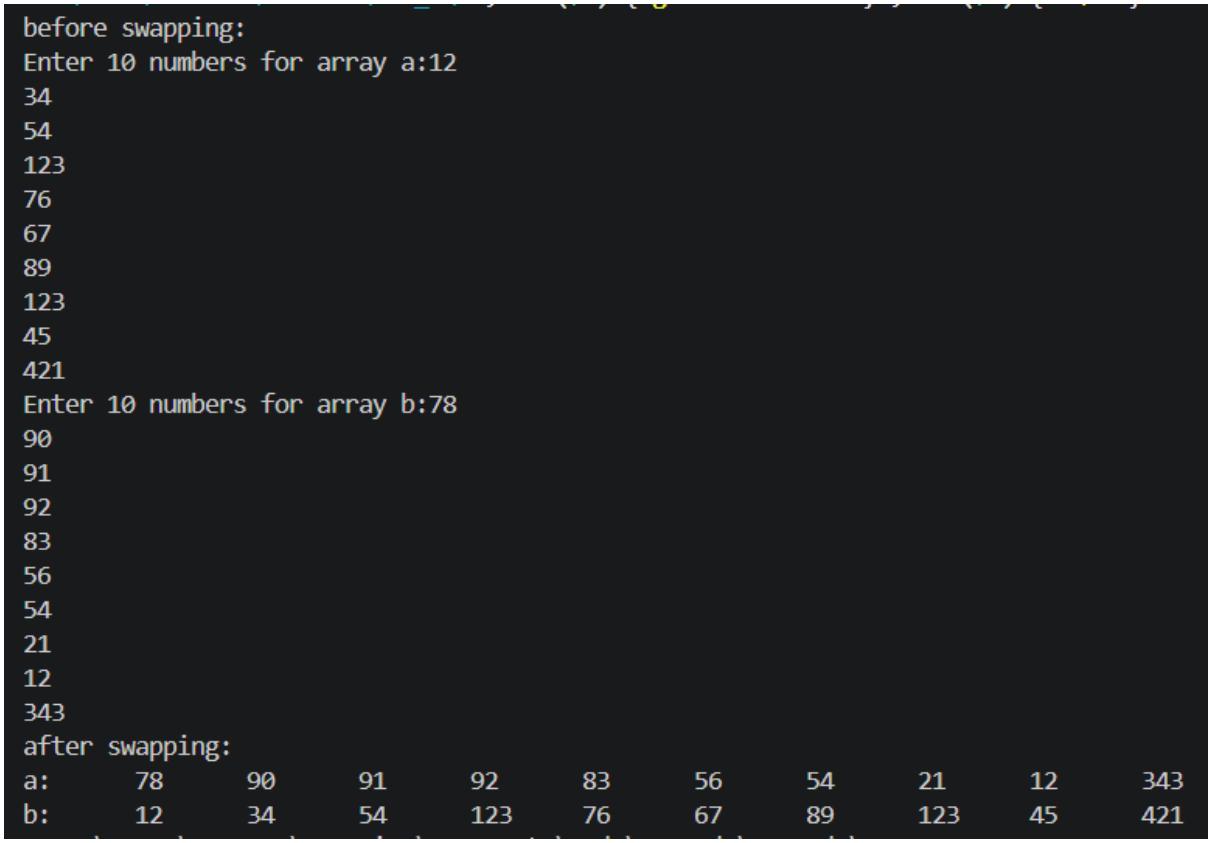


48) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND SWAP VALUES THESE ARRAYS.

```
le > LAB_3 > C 48.c > main()
#include<stdio.h>
int main()
{
    int a[10],b[10], c, i;
    printf("before swapping:\n");
    printf("Enter 10 numbers for array a:");
    for(i=0;i<10;i++)
        {
            scanf("%d", &a[i]);
        }

    printf("Enter 10 numbers for array b:");
    for(i=0;i<10;i++)
        {
    scanf("%d", &b[i]);
        }
    for(i=0;i<10;i++)
        {
            c=a[i];
            a[i]=b[i];
            b[i]=c;
        }
    printf("after swapping:\n");
    printf("a:\t");
    for(i=0;i<10;i++)
        {
                printf("%d\t", a[i]);
        }
    printf("\nb:\t");
    for(i=0;i<10;i++)
        {
            printf("%d\t", b[i]);
        }

    return 0;
}
```



49) PROGRAM TO SORT (SELECTION SORT) AN ARRAY.

```c
int n, i, j, min, temp;
int a[100];

printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter %d elements:\n", n);
for (i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}

for (i = 0; i < n - 1; i++)
{
    min = i;
    for (j = i + 1; j < n; j++)
    {
        if (a[j] < a[min])
        {
            min = j;
        }
    }
    if (min != i)
    {
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }
}

printf("Sorted array:\n");
for (i = 0; i < n; i++)
{
    printf("%d ", a[i]);
}
```

```
Enter number of elements: 6
Enter 6 elements:
3
4
1
6
5
8
Sorted array:
1 3 4 5 6 8
```

50) PROGRAM TO SORT (BUBBLE SORT) AN ARRAY.

```c
int main()
{
    int i, j, temp, size;
    printf("Enter the size of the array:");
    scanf("%d", &size);
    int n[size];
    printf("Enter the numbers:");4
    for(i=0;i<size;i++)
    {
        scanf("%d", &n[i]);
    }
    for(i=0;i<=size;i++)
    {
        for(j=0;j<size-1;j++)
        {
            if(n[j]>n[j+1])
            {
                temp=n[j];
                n[j]=n[j+1];
                n[j+1]=temp;
            }
        }
    }
    printf("sorted arrays:\t");
    for(i=0;i<size;i++)
    {
        printf("%d\t", n[i]);
    }
    return 0;

}
```

```
Enter the size of the array:5
Enter the numbers:3
5
8
7
9
sorted arrays:  3       5       7       8       9
```